

Software Requirements Specifications

For: playIT

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Change History

Version	Date	Description	Author
1.0	May 4, 2026	Initial draft of SRS based on Project Proposal (Revised 4)	Development Team
2.0	May 4, 2026	Comprehensive revision to strictly align with approved proposal details, including gamification mechanics, accurate constraints, and IEEE structural standards.	Requirements Engineer

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1. Introduction

1.1. Purpose

The purpose of this document is to provide a detailed, consistent, and verifiable description of the software requirements for **playIT**. playIT is a gamified English phonics mobile application with offline speech recognition and automated progress tracking. This document serves as the formal contract between stakeholders and the development team, establishing the basis for system design (SDD), implementation, and validation.

1.2. Scope

playIT is an Android-based mobile application designed for Grade 1 Filipino learners to support home-based independent reading practice. The system focuses on establishing foundational decoding skills through systematic English phonics instruction strictly aligned with the DepEd curriculum.

In-Scope:

- Instruction for letter sounds following the Marungko sequence: (m, s, a, i, o, b, e, u, t, k, l, y, n, g, p, r, d, h, w, c, f, j, q, v, x, z).
- Three core learning sub-levels per letter: "Hear It" (Audio Modeling), "Say It" (Offline Speech Recognition), and "Find It" (Interactive Picture Discrimination).
- Gamification mechanics including a localized heart-based penalty/recovery system, daily streaks (with unlockable badges), and a tiered star-rating system.
- An offline Parent/Guardian Progress Dashboard powered by a local SQLite/Room database, capable of PDF report generation.

Out-of-Scope:

- iOS support, web browser deployment, and cloud synchronization (the system is fully offline post-installation).

1.3. Definitions, Acronyms and Abbreviations

- **API:** Application Programming Interface.
- **DepEd:** Department of Education (Philippines).
- **Marungko Approach:** A reading sequence designed to enhance early reading skills by introducing letters based on the frequency of use rather than alphabetical order.
- **Vosk:** An offline speech recognition engine that supports multiple languages, including English, with optimized performance for mobile Android devices. It provides accurate phoneme-level recognition suitable for children's voices and accented English.
- **Room / SQLite:** A local database infrastructure used within Android to store user data entirely offline.
- **Phoneme:** The distinct units of sound in a specified language that distinguish one word

from another.

1.4. References

- Team 2526-sem2-it332-56 Software Project Proposal (REVISED 4).
- IEEE Guide for Software Requirements Specifications.
- DepEd Grade 1 English Curriculum Competencies (First to Second Quarter).

2. Overall Description

2.1. Product perspective

playIT is an independent, self-contained mobile application designed to operate completely offline after its initial installation. It bypasses the connectivity limitations typical of Philippine public school learners by utilizing local storage (Room/SQLite) for all user progression data and an integrated offline speech recognition library (Vosk) for vocal evaluation.

2.2. User characteristics

- **Primary Users (Grade 1 Learners):** Children aged 6-7 years old. They have minimal technical literacy, short attention spans, and require highly engaging, multi-sensory interfaces with immediate binary feedback.
- **Secondary Users (Parents/Guardians):** Adults responsible for supervising home-based practice. Technical literacy varies from low to moderate. They require simplified dashboards and exported PDF reports to easily interpret their child's progress without specialized educational knowledge.

2.3. Constraints

- **Platform Restriction:** The application is constrained to the Android Operating System, requiring a minimum of Android 8.0 (API Level 26) to ensure compatibility with older devices.
- **Hardware Limitations:** Accurate speech recognition heavily depends on device microphone quality and acoustic properties.
- **Data Storage:** All user progression data, streaks, and generated reports must be stored entirely locally on the device.
- **Speech Recognition Limitations:** Vosk has inherent technical limitations processing child voices; therefore, the system is constrained to a baseline minimum $\geq 75\%$ detection accuracy target to accommodate Filipino-accented English phonemes and high-pitched vocal ranges.

2.4. Assumptions and dependencies

- Users possess a functioning Android smartphone or tablet with microphone and speaker access.
- An internet connection is available *only once* for the initial downloading of the APK and the Vosk language models.

- The home-based learning environment maintains ambient noise levels of $\leq 40\text{dB}$ to ensure the speech recognition engine functions accurately.

3. Specific Requirements

3.1. External interface requirements

3.1.1. Hardware interfaces

- **Microphone:** The system shall interface with the device's built-in microphone to capture user vocalizations for the "Say It" module.
- **Audio Output:** The system shall utilize the device's speakers or headphones to deliver pre-recorded native speaker phonemes. The auditory reference tones must maintain high fidelity, with a maximum frequency deviation of ± 10 cents.
- **Touchscreen:** The system shall require a standard touchscreen to interact with the gamified elements and navigation grid.

3.1.2. Software interfaces

- **Android OS:** The system shall execute on Android API level 26 or higher.
- **Vosk Library:** The system shall integrate the Vosk offline speech recognition library (minimum version 0.3.45) for local audio processing and phoneme decoding. The English small model shall be bundled with the APK.
- **SQLite/Room Database:** The application shall utilize Android Room to execute CRUD operations on the local SQLite database for telemetry, streak tracking, and reporting.

3.1.3. Communications interfaces

- No runtime communication interfaces are required. Once configured, all data processing, including speech evaluation and report generation, occurs 100% offline.

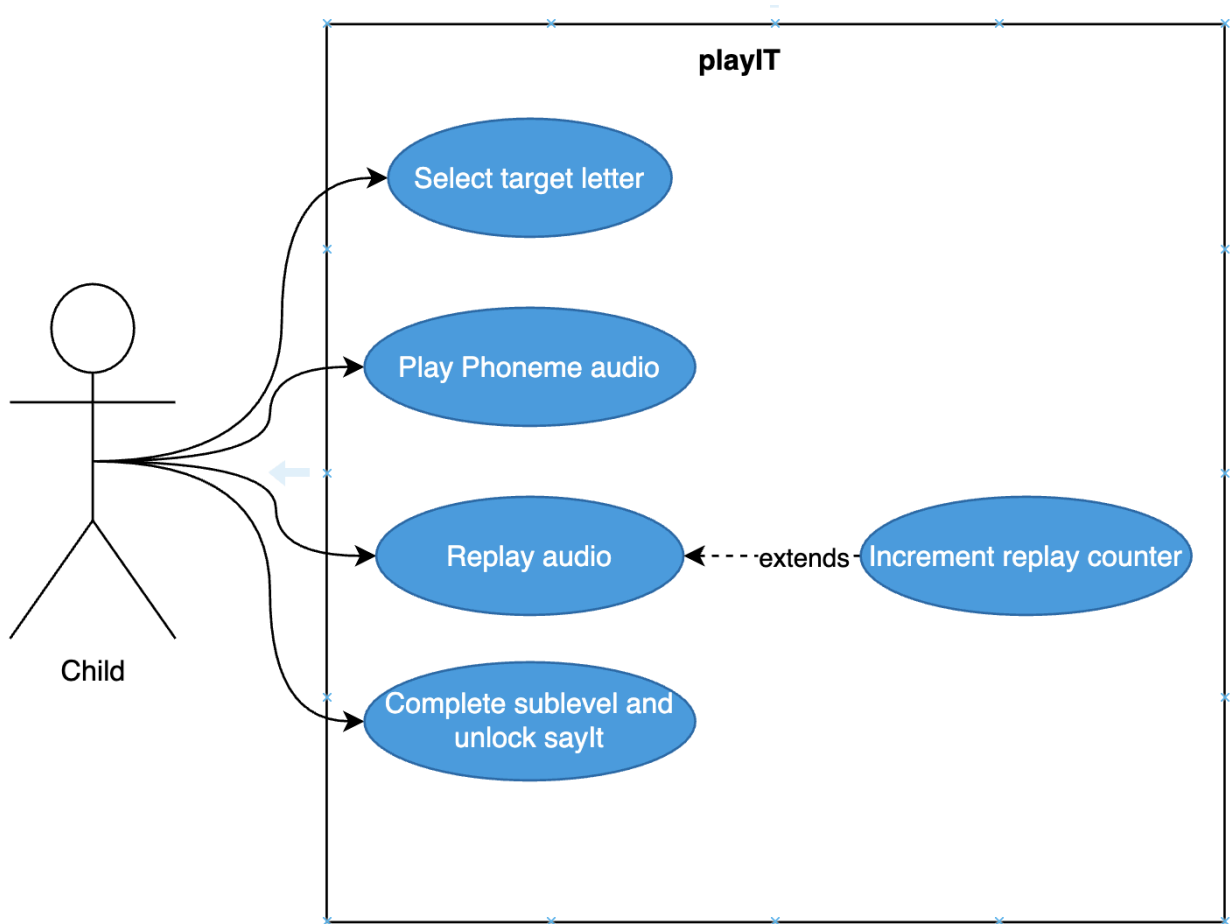
3.2. Functional requirements

Module 1: Letter-Sound Acquisition ("Hear It")

1.1 Playback Phoneme Audio Model

- **Use Case Description:** The system delivers a 100% accurate phonemic model of the target letter using pre-recorded native English speaker audio, accompanied by an animation.
- **Pre-condition:** The user has successfully unlocked the target letter in the sequence.
- **Main Flow:** 1. User selects the target letter and taps the "Hear It" sub-level. 2. System loads the specific pre-recorded audio file. 3. System plays the audio (e.g., "/mmm/" for M like Mouse") and renders an animated visual of the letter alongside a representative picture. 4. User taps "Next" to complete the sub-level.
- **Alternative Flow:** User replays the audio multiple times before clicking "Next". The system increments a "replay count" metric.
- **Post-condition:** The "Say It" sub-level is unlocked for the current letter.

- Use case diagram

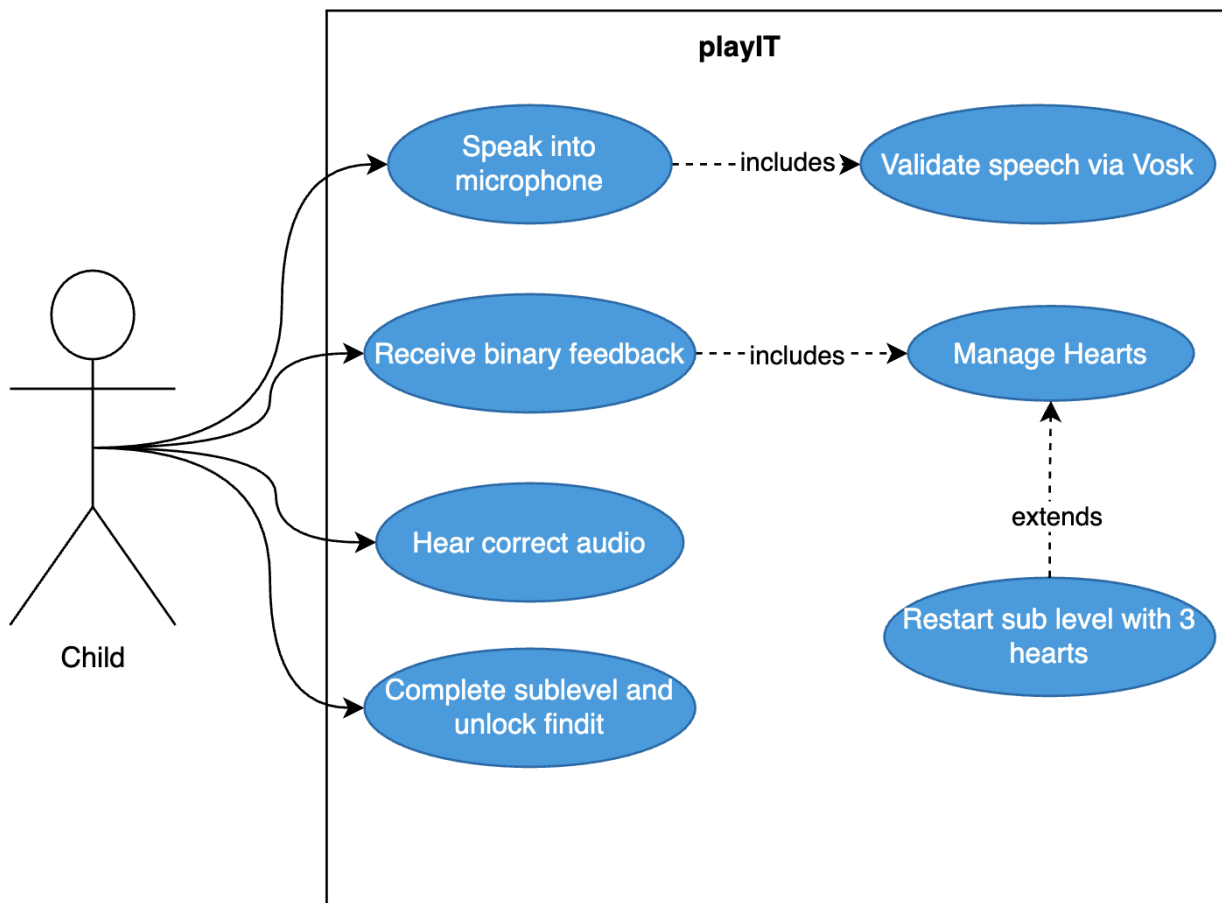


Module 2: Speech Production ("Say It")

2.1 Process Speech Input and Provide Binary Feedback

- **Use Case Description:** The system captures the user's vocal pronunciation, processes it locally via Vosk, and enforces exactly 1 correct production to pass.
- **Pre-condition:** User has completed the "Hear It" module for the current letter and possesses an initial pool of 5 hearts.
- **Main Flow:**
 1. System prompts the user to speak.
 2. User speaks into the device microphone.
 3. System analyzes the frequency and compares it to the native phoneme reference.
 4. If the phoneme matches (1 correct production required): System triggers a green visual highlight, plays a success sound, awards the sub-level pass, and proceeds.
- **Alternative Flow (Incorrect Input / Heart Deduction):** If the speech is unmatched, the system highlights red, plays corrective audio ("/m/"), and deducts 1 heart from the pool. The user is prompted to retry (up to 5 attempts).

- **Alternative Flow (Heart Depletion):** If the user fails 5 times (hearts deplete to 0), the system automatically restarts the current sub-level and initializes the player with 3 hearts instead of 5.
- **Post-condition:** Metrics (success/fail, attempts, hearts lost) are securely saved to the SQLite database. The "Find It" module is unlocked.
- **Use-case diagram**



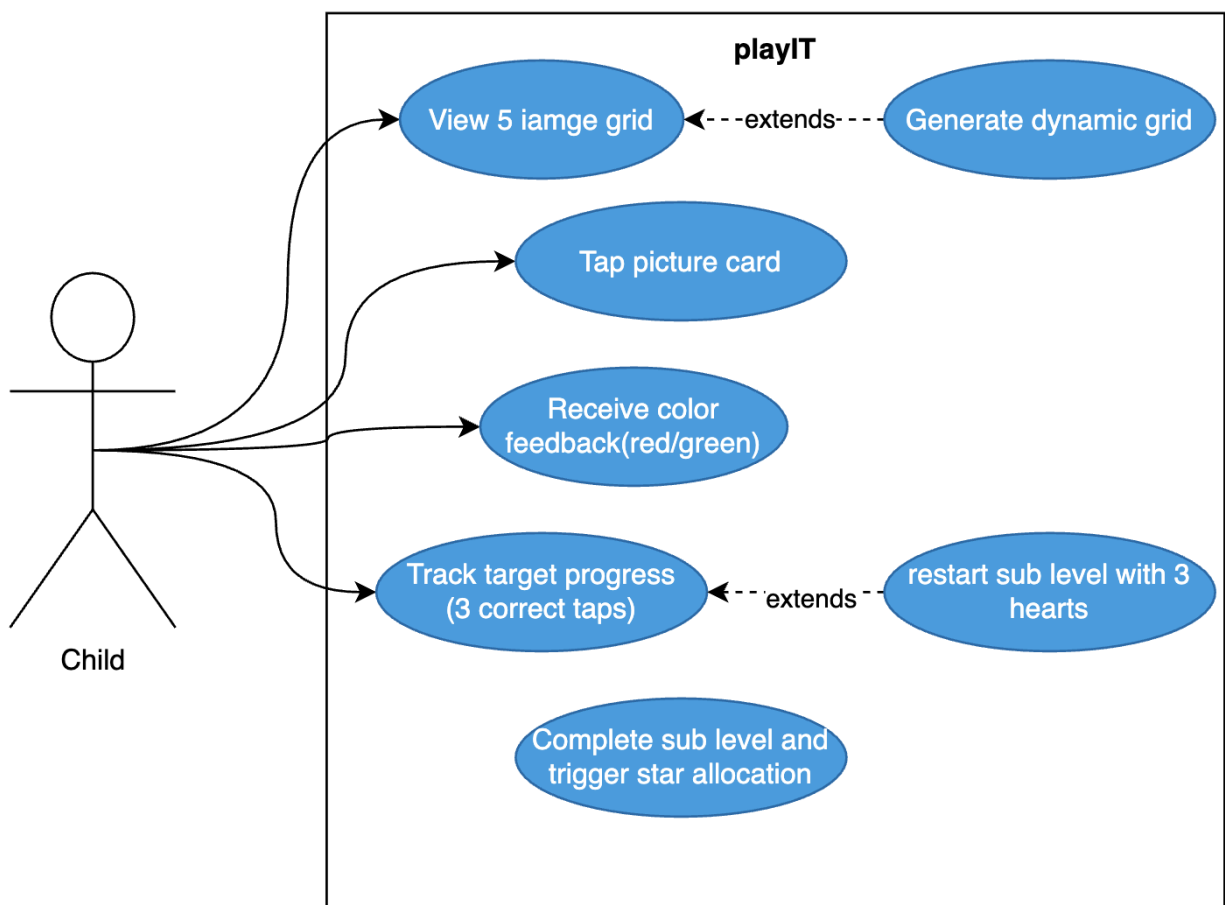
Module 3: Phoneme Discrimination ("Find It")

3.1 Evaluate Dynamic Grid Selection

- **Use Case Description:** The system assesses visual-auditory discrimination by presenting a dynamic 5-image grid. Exactly 3 pictures represent the target sound, while 2 distractor pictures are sourced specifically from *previously mastered letters*.
- **Pre-condition:** User has passed the "Say It" module for the current letter. The heart pool is set to 5.
- **Main Flow:**
 1. System generates a 5-image grid (3 targets, 2 distractors from past letters).
 2. User taps an image.
 3. If correct: The image highlights green, a success sound plays, and the internal score

increments.

4. If incorrect: The image highlights red, a failure sound plays, the image disables, and 1 heart is deducted.
 5. The process repeats until the user achieves 100% discrimination accuracy for the round (successfully tapping all 3 correct images).
- **Alternative Flow (Heart Depletion):** If hearts reach 0 before finding all 3 targets, the sub-level instantly restarts. The grid is randomly reshuffled, and the user's heart pool is reset to 3.
 - **Post-condition:** The letter is marked as fully completed. Overall letter metrics are submitted to the local database, and star allocation is triggered.
 - **Use case diagram**



Module 4: Word Construction (“Blend IT”)

- **Use Case Description:** After completing all letters in a Marungko group, the system unlocks a Blend It checkpoint node. The child is presented with a picture and hears the target word automatically. A shuffled bank of letter tiles (correct letters + 1–3 distractors) is provided. The child taps tiles to fill letter slots in correct order, then submits. After 2 wrong attempts, a hint locks one correct tile into place automatically.
- **Pre-condition:** All letters in a group (e.g., Group 1: m, s, a, i) are fully completed. The user taps an unlocked Blend It node on the Map Screen. The user has a fresh pool of 5

hearts.

- **Main Flow:**

- System loads a session of 5 randomly selected words (at least 1 from the current group, others from previously mastered groups).
- For each word:
 - System displays a picture of the target word and automatically plays its audio.
 - System renders empty letter slots matching the word length.
 - System displays a shuffled tile bank containing the correct letters plus 1–3 distractor letters.
 - User taps letter tiles to fill slots in correct order (left to right).
 - User taps a filled slot to remove a letter back to the tile bank.
- When all slots are filled, the Submit button activates.
- User taps Submit:
 - **If correct:** System plays success chime, displays green feedback, and advances to the next word.
 - **If incorrect:** System plays buzz sound, displays red feedback, deducts 1 heart, and resets the word for retry.

Alternative Flow (Hint Trigger):

- After 2 wrong attempts on the same word, the system automatically locks one correct letter tile into the correct slot position.

Alternative Flow (Heart Depletion):

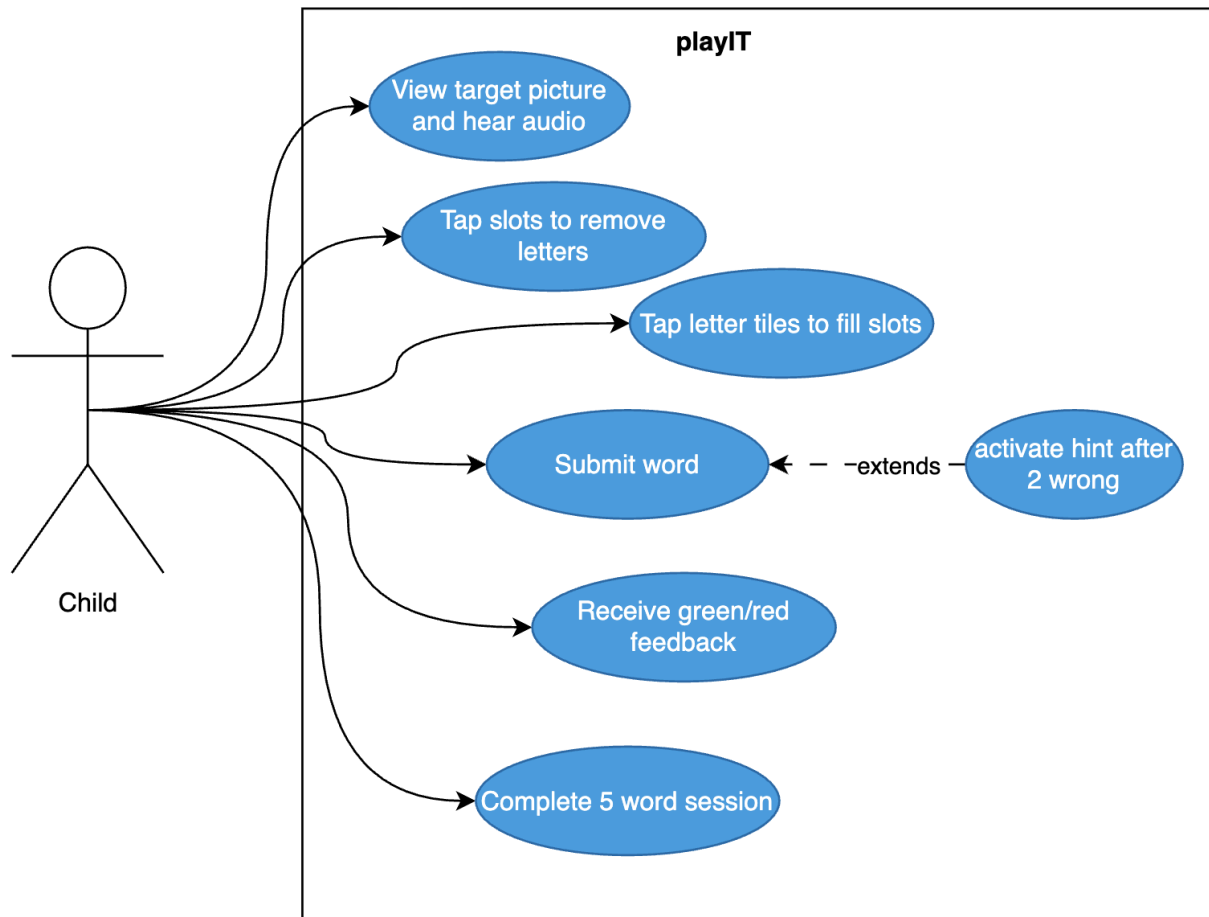
- If hearts reach 0 before completing all 5 words, the session ends immediately with no stars awarded.

Alternative Flow (Audio Replay):

- User can tap the picture at any time to replay the target word audio.

Post-condition: After completing all 5 words, the system displays a star rating (1–3 stars) based on words correct and hearts lost. The Blend It node on the Map Screen updates with earned stars. Progress is saved locally to the device

- **Use case diagram.**



Module 5: Mastery Progression & Gamification

4.1 Calculate Star Rating and Heart Recovery

- **Use Case Description:** The system evaluates aggregate sub-level performance to grant stars and manages positive reinforcement through heart recovery.
- **Pre-condition:** The user completes all three sub-levels ("Hear It", "Say It", "Find It") for a given letter.
- **Main Flow:**
 1. System applies star logic:
 - 3 Stars = 100% accuracy and 0 hearts lost.
 - 2 Stars = $\geq 80\%$ accuracy.
 - 1 Star = Passed, but with lower accuracy/multiple hearts lost.
 2. The system executes the heart recovery rule: The system awards +1 heart for every 3 consecutive correct answers achieved across interactive modules.
- **Post-condition:** Star ratings update the UI.

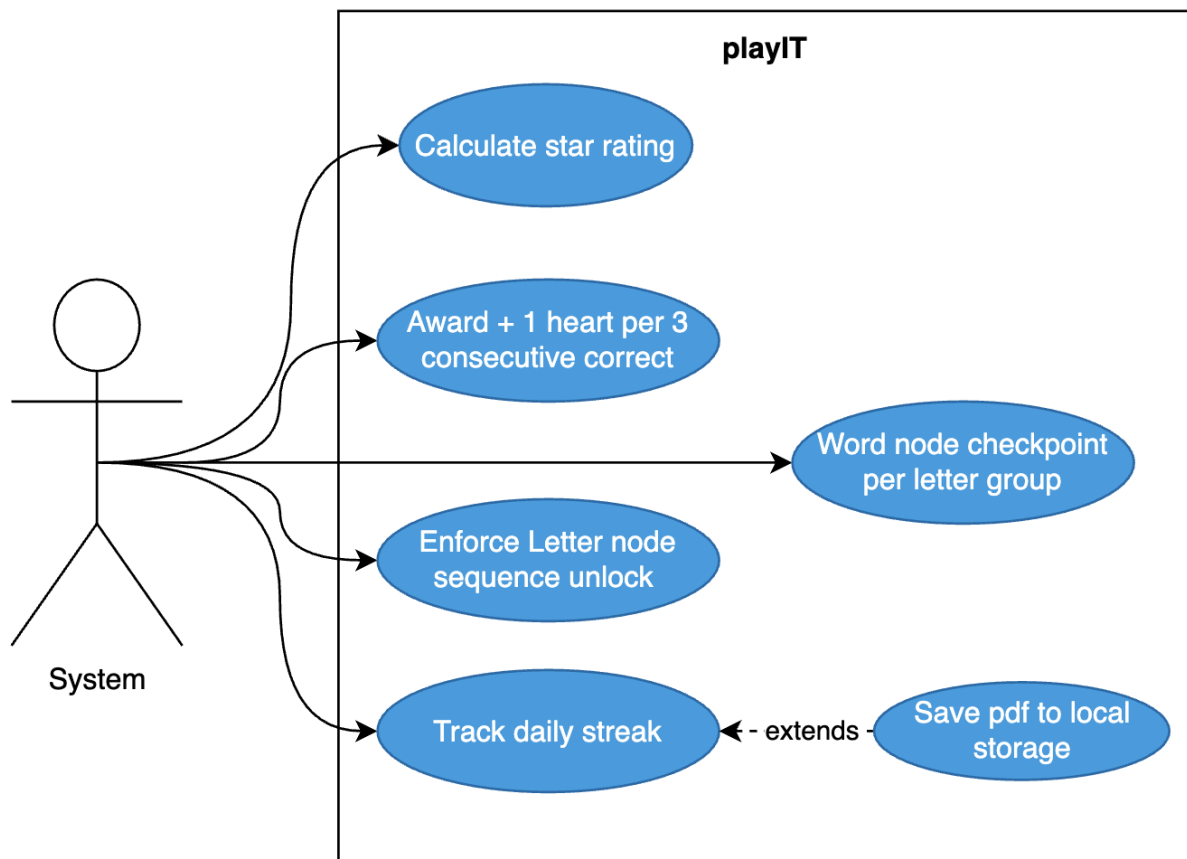
4.2 Manage Marungko Unlock Progression

- **Use Case Description:** The system rigorously enforces the complete 28-letter Marungko sequence.
- **Pre-condition:** The application boots into the Level Map UI.

- **Main Flow:** The system evaluates completion states. Letter N+1 remains locked and inaccessible until all three sub-levels of Letter N are passed. The strict sequence enforced is: m, s, a, i, o, b, e, u, t, k, l, y, n, g, ng, p, r, d, h, w, c, f, j, ñ, q, v, x, z.

4.3 Manage Learning Streaks

- **Use Case Description:** The system encourages continuous practice by tracking consecutive days of activity.
- **Pre-condition:** User logs into the application and completes at least one module interaction.
- **Main Flow:** 1. The system updates the consecutive login counter. 2. If the user hits milestone days (5, 10, 15, or 20 consecutive days), the system automatically unlocks and displays the corresponding Streak Badge.
- **Alternative Flow:** If 24 hours pass with zero user activity, the system resets the streak counter to 0.
- **Use case diagram**



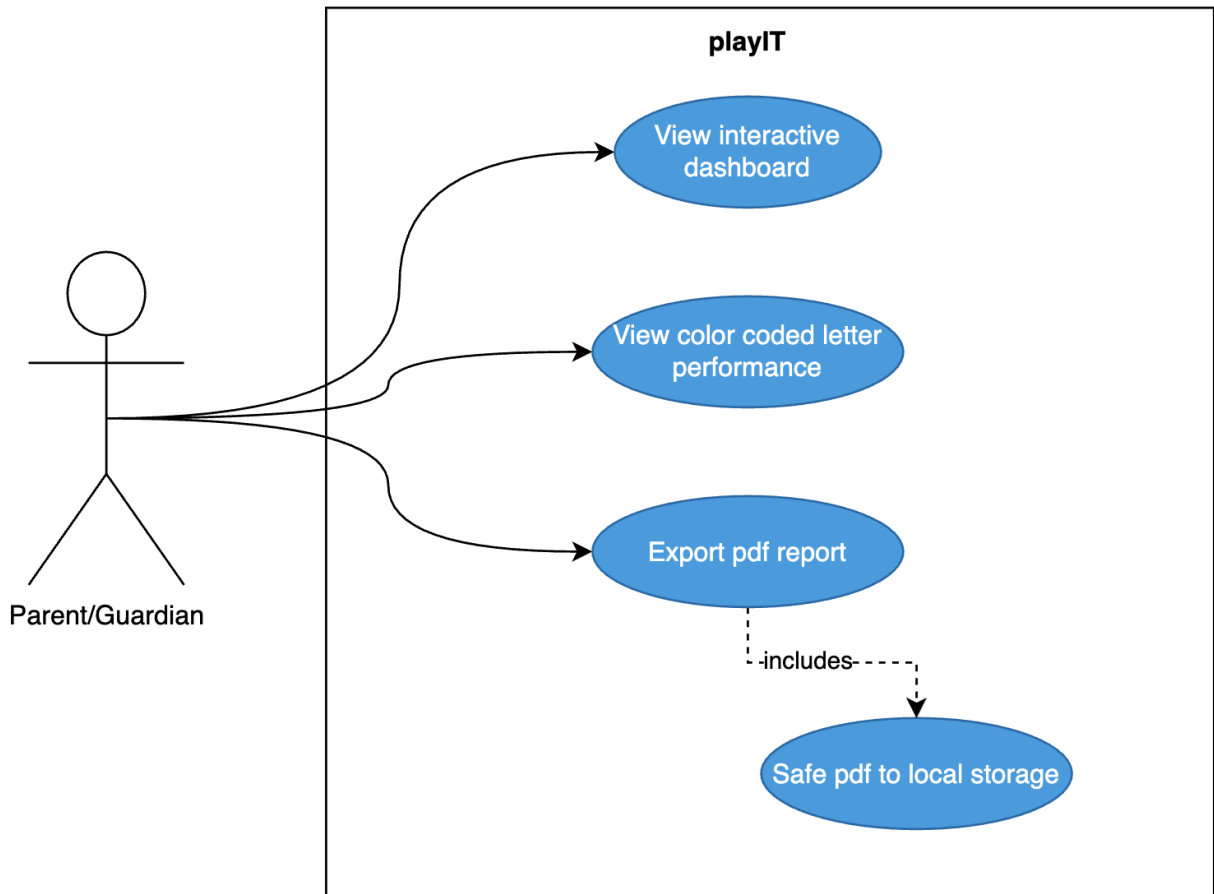
Module 6: Parent/Guardian Dashboard

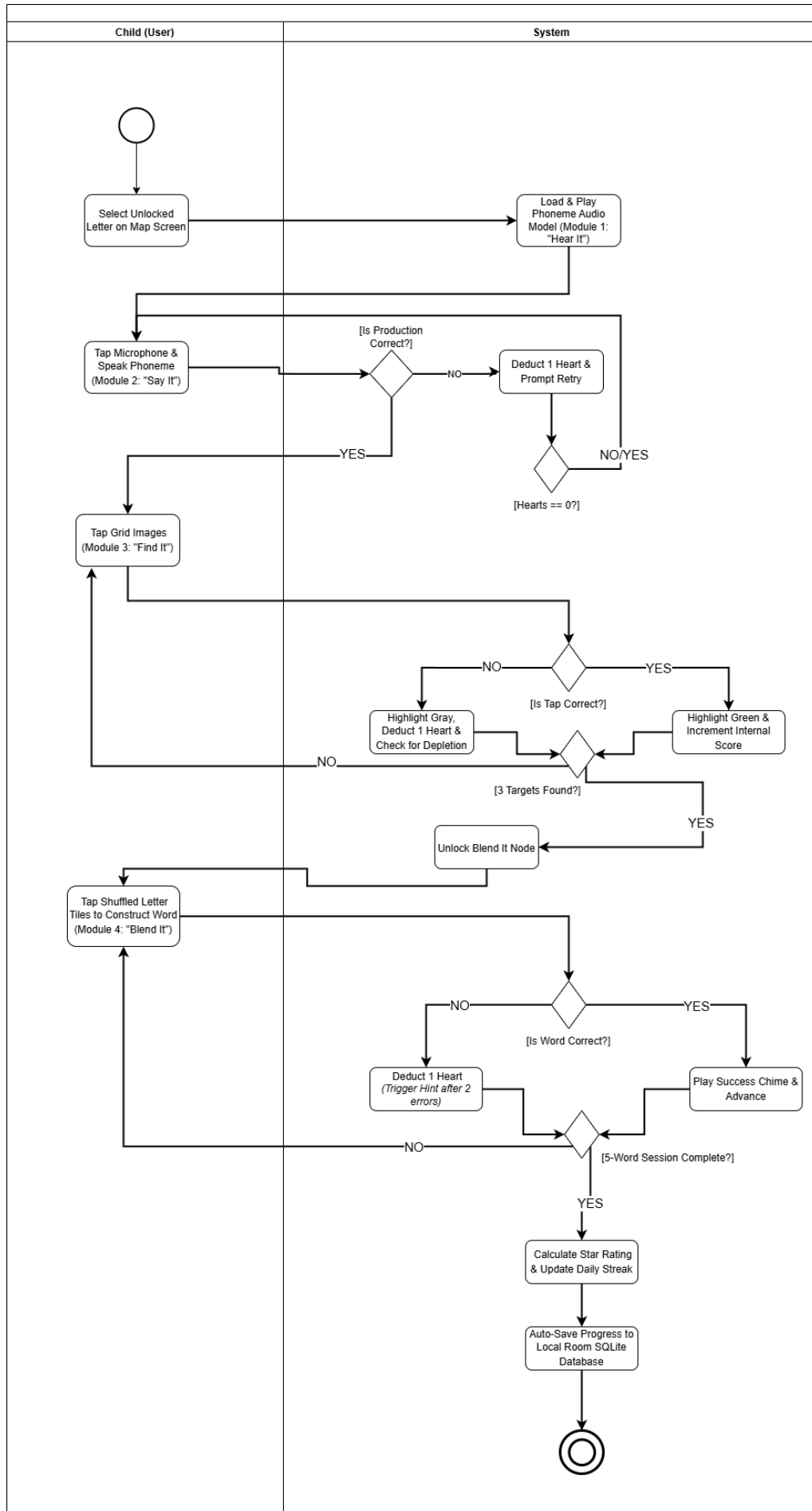
6.1 Present Interactive Progress Dashboard

- **Use Case Description:** The system aggregates metrics into an analytical view for parents to review performance and monitor their child's progress.
- **Pre-condition:** The app is open and the user is on the main menu.
- **Main Flow:**
 1. User taps the "Parent Dashboard" button in the main menu
 2. System immediately displays the dashboard (no login required)
 3. System queries the SQLite database
 4. System renders the following offline metrics:
 - Accuracy trend (%) per letter
 - Attempts per letter
 - Hearts lost per letter
 - Time to complete per letter
 - Retention scores (last 7 days)
 - Stars earned per letter
 5. System color-codes letters:
 - Green = $\geq 80\%$ accuracy (mastered)
 - Yellow = 50-79% accuracy (developing)
 - Red = $< 50\%$ accuracy or 3+ failed attempts (at-risk)
- **Post-condition:** Parent views the data and can choose to export a PDF report.

6.2 Generate and Export PDF Reports

- **Use Case Description:** The system compiles the dashboard data into a formalized PDF document for external viewing or printing.
- **Pre-condition:** The parent is inside the dashboard.
- **Main Flow:** 1. Parent clicks the "Export Report" button. 2. System automatically converts the current data views (metrics and at-risk flags) into a formatted PDF document. 3. System saves the PDF directly to the device's local storage.
- **Post-condition:** The PDF file is available locally on the device for the parent to share or print.
- **Use case diagram**





3.3. Non-functional requirements

3.3.1. Performance

- **Audio Loading:** The system shall process each letter's audio model ("Hear It") within a maximum duration of ≤ 5.0 seconds.
- **Speech Recognition Latency:** The system shall process user vocal input and display immediate binary feedback within ≤ 0.5 seconds of microphone deactivation.
- **UI Responsiveness:** Visual feedback mapping to touch inputs (e.g., green/red highlights in "Find It") must render within ≤ 0.3 seconds.

3.3.2. Security

- **Data Privacy:** 100% of telemetry, user profiles, and vocal recordings shall be processed and stored strictly on the local device via SQLite. No data shall ever be transmitted externally.
- **Access Control:** The application does not require user accounts or logins.

All functionality is accessible from the main menu. Destructive actions (reset progress, delete data) shall require solving a simple arithmetic problem to prevent accidental activation by children.

3.3.3. Reliability

- **Speech Detection Threshold:** To account for the highly variable nature of child voices and acoustic variations, the Vosk engine shall be technically calibrated to achieve a minimum baseline detection accuracy of $\geq 75\%$. (Note: While system detection allows a 75% baseline to prevent undue frustration, pedagogical mastery for a 2-star rating remains strictly at $\geq 80\%$ accuracy).
- **Data Persistence:** The system shall continuously execute auto-saves to the local Room database upon the completion of every sub-level interaction to prevent data loss.

3.3.4. Environmental Resilience

- **Acoustic Monitoring:** The system shall feature an in-app noise level indicator that visually alerts the user/parent if ambient environmental noise exceeds the recommended ≤ 40 dB threshold required for accurate speech recognition.

4. Requirements Traceability Matrix

Req ID	Requirement Name	Source / Proposal Feature	Testing/Validation Target
FR-01	Playback Phoneme Audio	Gen Obj 1 / Module 1	TC-01: Verify audio load ≤ 5 s, pitch

			deviation $\leq \pm 10$ cents.
FR-02	Process Speech Input	Gen Obj 2 / Module 2	TC-02: Verify Vosk latency $\leq 0.5s$ & baseline accuracy $\geq 75\%$.
FR-03	Enforce Pass Condition	Gen Obj 2 / Module 2	TC-03: Verify exactly 1 correct vocal production passes "Say It".
FR-04	Heart Depletion & Reset	Gen Obj 2 & 3 / Mechanics	TC-04: Verify 5 start hearts, -1 per error, restart at 3 hearts on 0.
FR-05	Grid Evaluation & Distractors	Gen Obj 3 / Module 3	TC-05: Verify 3 targets, 2 distractors (from mastered letters only).
FR-06	Star Calculation Rules	Feature 2 / Gamification	TC-06: Verify 1/2/3 star distribution based on hearts & accuracy.
FR-07	Heart Recovery System	Feature 2 / Gamification	TC-07: Verify +1 heart is awarded per 3 consecutive correct taps.
FR-08	Marungko Lock Progression	Feature 3 / Gamification	TC-08: Verify Letter N+1 remains locked until Letter N passes all 3 parts.
FR-09	Learning Streaks & Badges	Feature 4 / Gamification	TC-09: Verify badge unlock at 5/10/15/20 days & 24h inactivity reset.

FR-10	Dashboard Metrics Display	Feature 5 / Dashboard	TC-11: Verify accurate local display of all metrics (incl. retention score).
FR-11	At-Risk Letter Flags	Feature 6 / Dashboard	TC-12: Verify Green/Yellow/Red flag thresholds apply properly.
FR-12	PDF Report Export	Parent Dashboard Specs	TC-13: Verify system writes formatted PDF progress report to local storage.